



EVERETT-ALAN HOOD

# SURROUND SOUND V2

# Motivation & Background

Auditory AI is underexplored outside speech and music.

Prior work: DCASE Challenge separates ASC and SED; this project combines them into one live system.

V1 (ML class) proved the concept with lightweight CNNs {macro-F1 ~0.50 env, ~0.39 events}

Key failures pointed the way forward:

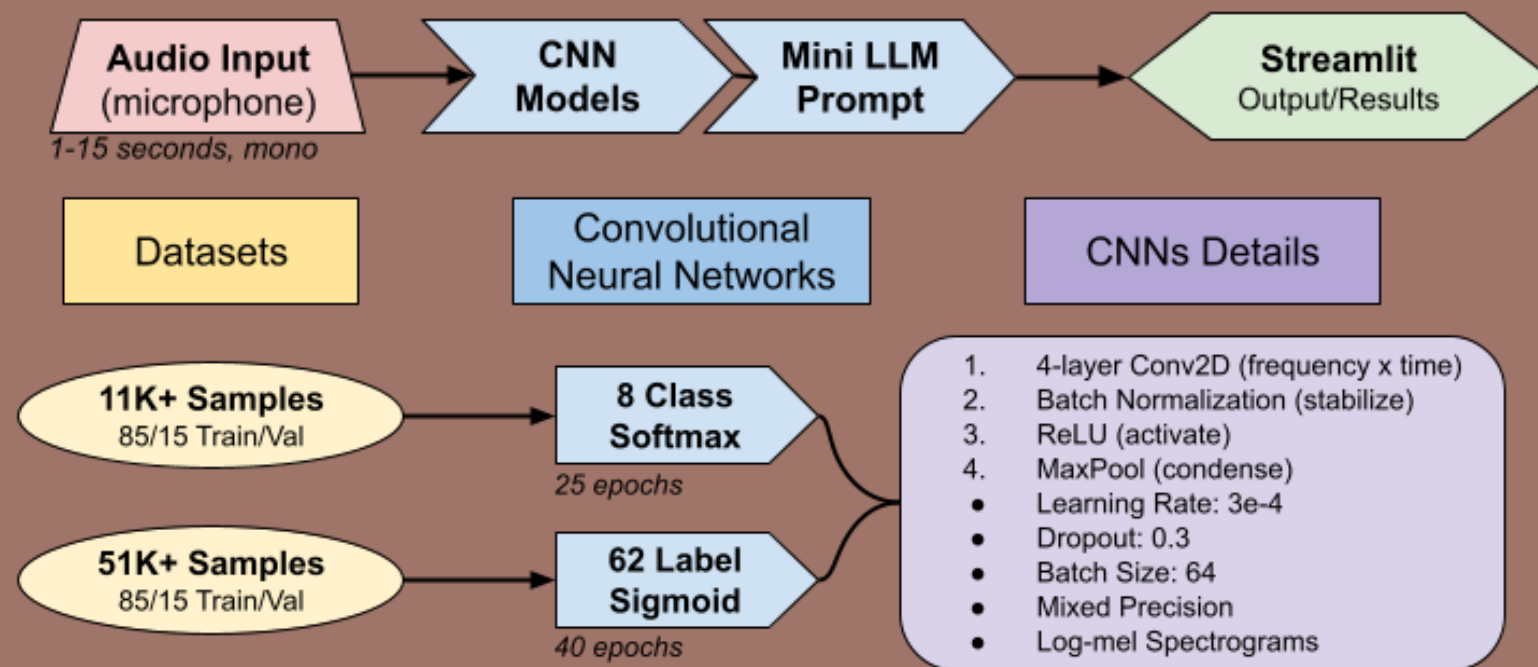
- class imbalance
- acoustic overlap
- limited temporal context
- weak LLM integration





# Proposed Approach

dual-head CNN (env classifier + event detector) from V1

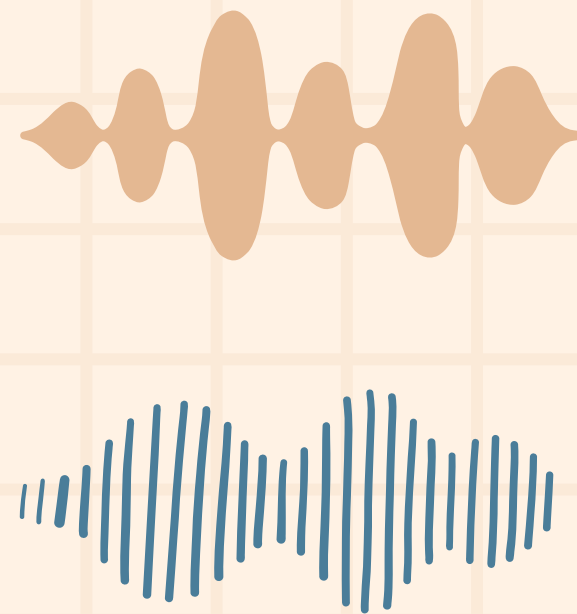


# Upgrades

- Adding a Speech Layer to the existing Pipeline
- Deeper / more expressive architecture per head (residual blocks, attention layers)
- Better handling of long-tail class imbalance (focal loss, mixup augmentation)
- Temporal modeling: moving toward streaming-aware inference rather than fixed 10s windows
- Cleaner LLM summarization as a proper output stage, not a post-hoc add-on
- Any beneficial concepts we learn in this class

## Use Cases & Applications

- Accessibility & Assistance
- Social Awareness
- Noise Pollution Detection
- Autonomous Robotics
- Track Behavioral Changes
- Emergency Response



## DL 586 Objective

- Create a model that classifies a user's environment from real-time audio.
- Predict likely outcomes of the current scenario based on surrounding context.

# Rough Idea

- Clean raw audio & metadata of the human experience
- Use embeddings to train multi-dimensional classification system
- Record in real time from the user's microphone
- Predict the current scenario

## Datasets

- AudioSet (environment, 8 classes, ~11K samples)
- FSD50K (events, 62 labels, ~51K samples)
- Speech Dataset?

HuggingFace, Zenodo, Kaggle

### Attributes/Features

- WAVFormat
- Mono
- 16kHz
- normalized 10s segments
- 128-band log-mel spectrograms
- LUFS & DB



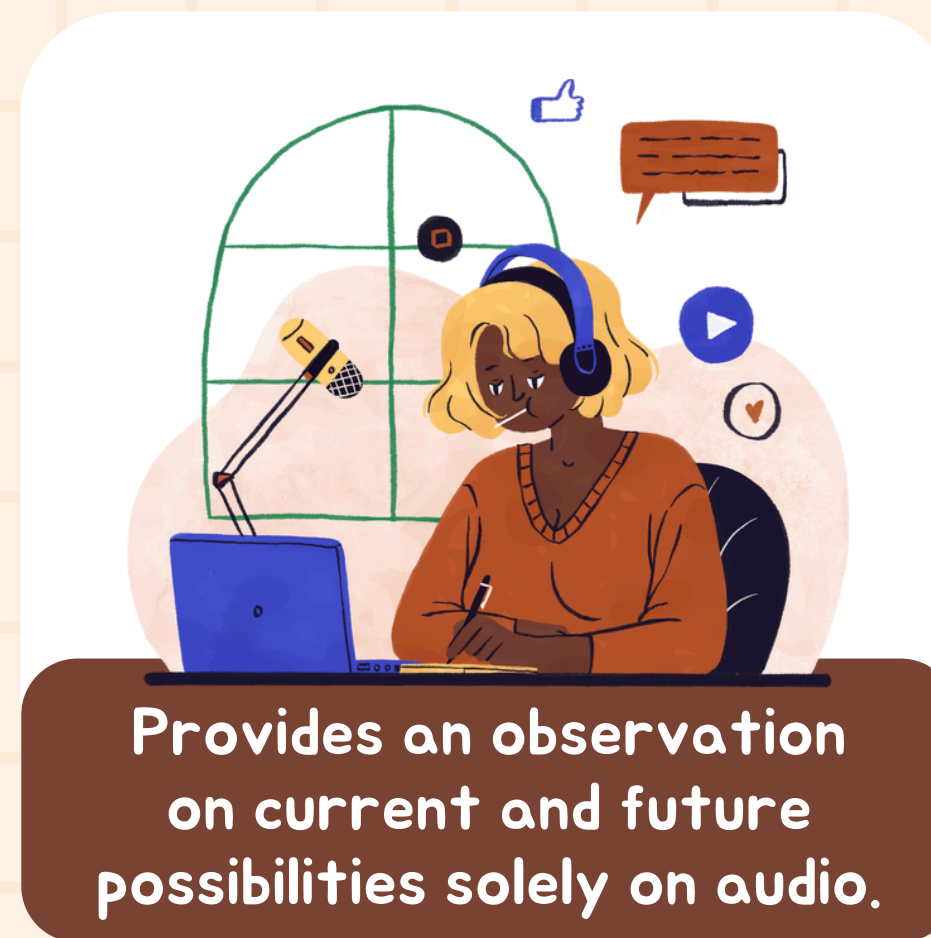
### Sample Example

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{  
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  "sample_rate": 16000,  
  "channels": 1,  
  "duration_s": 10.0,  
  "spectrogram_shape": [1, 128, 313],  
  "environment_label":  
    "outdoor_nature",  
  "event_labels": ["Wind", "Bird",  
    "Flowing Water"],  
  "split": "train"  
}
```





# Audio is info. Information is revelation.



We want our programs to cooperate with us and keep us safe.  
Why not put technology in our shoes?

**THANK YOU!**